

GENREG / EEC

Audit & Re-test of the Cognition Findings

Hunting holes in the logic; re-testing every load-bearing claim
with high seed counts; separating what emerged from what was shaped.

Central correction: "no gradients ever" bans designed reward-gradients,
not just gradient optimisers. Several headline results leaned on shaped rewards.

1. Executive summary

What this audit did

Re-read every claim in the cognition arc (FINDINGS.md A-K), listed the holes (AUDIT.md), then re-tested the load-bearing ones with 5-8 seeds and proper controls. The trigger was the rule 'no gradients ever' -- which, taken fully, forbids DESIGNED reward-gradients (reward shaping), not only gradient optimisers. That reframes the whole 'graded survival' story.

Headline corrections

HELD UP: external memory is invented and pays in real currency -- ablation 0.67 vs 0.16 (n=8), and in a pure-food foraging world 4.01 vs 1.93 on the same organism. Sexual reproduction robustly protects a shared language under noise (0.79 vs 0.36 at high noise, n=8). Word order is load-bearing (scramble collapses it, 0.22->0.055, n=8).

REPAIRED: 'sequential vastly beats bag' was a TRAINING ARTIFACT -- the bag was undertrained; at fair training the gap shrinks to a modest 0.37 vs 0.25. 'Memory never forgets' is refined: a one-step drop then a flat plateau to delay 24.

FAILED / SHAPED: content-addressing is a hard wall (fails three ways). The binding and spatial-layout results lean on DESIGNED graded rewards and are downgraded to reward-dependent. Compositionality is largely tautological (per-word grounding). The foraging 'relocating' control was confounded by abundance. One probe used an explicit designed gradient and is disavowed.

The refined thesis

Capabilities emerge from worlds whose SURVIVAL PHYSICS make them the cheapest way to live -- external memory most cleanly. Word order emerges too, but only when partial communication carries partial survival value (a realistic condition), and robustly across reward shapes. The lever is NOT 'add a gradient to the reward'; where we did that, the result is weaker.

2. The reward-shaping reckoning (the deepest hole)

The hole

The session's 'reachability lever' was: grade survival so partial progress pays. In three results that grading was something I DESIGNED, not a world-consequence: syntax (+1 per decoded role), binding (I switched the reward to per-entity until binding appeared), and spatial layout (per-position credit). Designing a slope toward the target is reward shaping -- a gradient in disguise -- even with a gradient-free optimiser.

The test

Re-ran the relational (syntax) world under five fitness SHAPES, 5 seeds each, asking whether order still emerges and stays load-bearing (Figure 1, left):

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all-or-nothing (whole message or death) : full 0.011 -> does NOT emerge (chance)
any (>=1 role right)                   : full 0.062 -> does NOT emerge
threshold2 (>=2 right, coarse)         : full 0.146 -> EMERGES, load-bearing
per_role (linear count)                  : full 0.225 -> EMERGES, load-bearing
square (convex)                         : full 0.228 -> EMERGES, load-bearing
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What it means

Order emerges across THREE different graded shapes but NOT under all-or-nothing. Robustness across shapes is the signature that the WORLD STRUCTURE (relations) drives it, not a tuned slope. So the result survives -- with a stated condition.

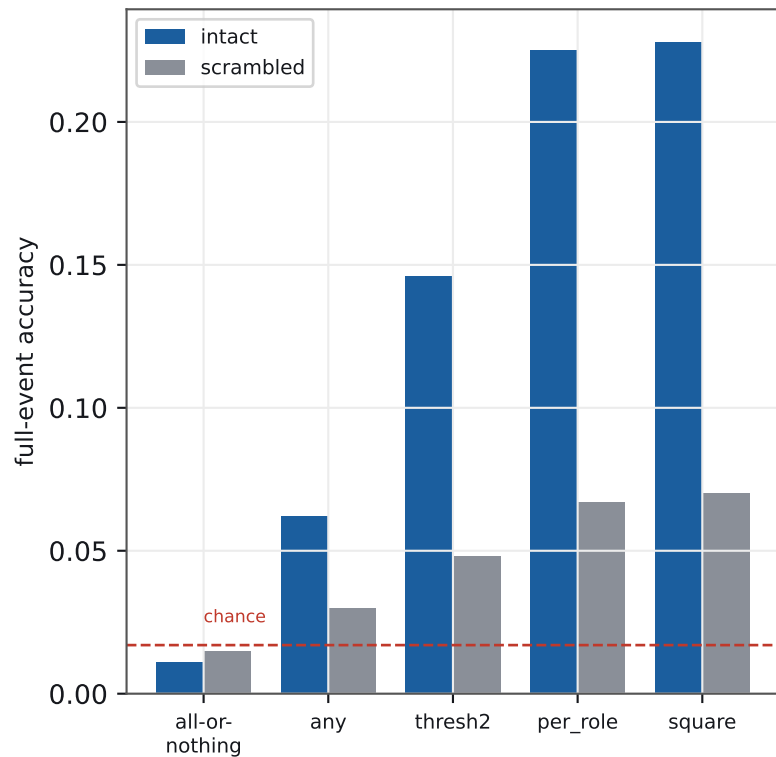
The condition: emergence requires partial communication to carry partial survival value. That is realistic (understanding 'predator!' helps even if you miss 'from the north'); all-or-nothing is the unnatural case. But it is NOT free -- in a world where only perfect comprehension keeps a partner alive, order is unreachable. 'Graded reward is a free lever' was the wrong lesson; 'partial success must carry partial survival value, as it does in real worlds' is the right one.

Reclassification

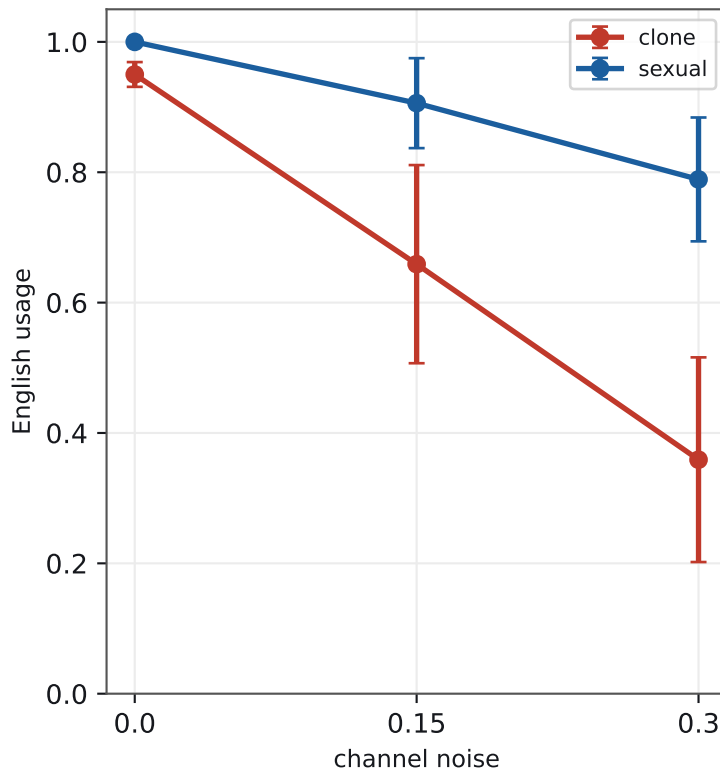
NATURAL grading (legitimate): food eaten (foraging), correct-recall count (memory), partial-comprehension value (syntax, robust to shape). DESIGNED grading (shaped, downgrade): the per-entity switch for binding; the killed proximity probe; 'make guessing worthless to open a gradient' in content-addressing.

Figure 1. Reward-shape dependence (left) and a result that held up (right)

Order emerges only with partial-value reward
(and is robust to its shape)



Sexual reproduction protects English
(n=8; clone has higher variance)



3. What held up under scrutiny

External memory is invented and load-bearing -- the strongest result

Invented (feedforward organism + persistent world slot): recall 0.67 +/- 0.19 with the slot vs 0.16 +/- 0.01 with it disabled (n=8). The high variance is seed-dependent REACHABILITY (some seeds never find the trick), but where it is found, severing the world collapses it to chance -- the memory is genuinely external.

And it pays in REAL CURRENCY: in a pure-food foraging world (fitness = food eaten, no shaping of any kind), the same evolved organism eats 4.01 with its marks vs 1.93 when marks are disabled -- a 2x effect. For a local-vision feedforward forager, marks can ONLY serve as external memory, so this is memory paying its way in survival. (Figure 2.)

Sexual reproduction stabilises a shared language

Under channel noise, cloning lets English erode (0.36 +/- 0.16 at noise 0.3) while sexual reproduction holds it (0.79 +/- 0.10). Crucially cloning also has ~1.6x the variance: it propagates whatever each lineage drifted to, while crossover regresses to the shared code. Mechanism and magnitude both robust at n=8.

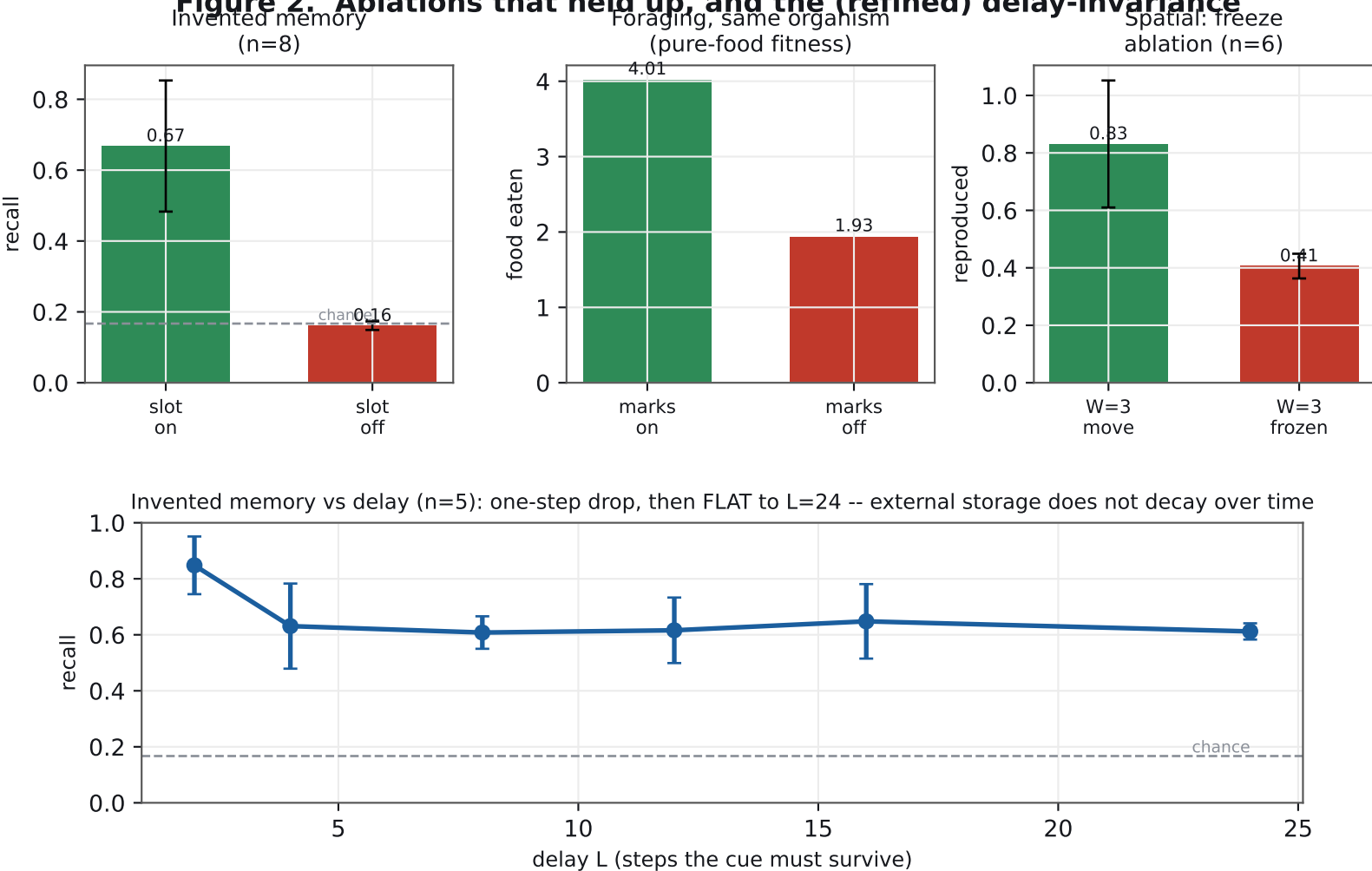
Word order is genuinely used (not just present)

Scrambling the message slots before decoding collapses full-event accuracy from 0.22 to 0.055 (n=8). Meaning lives in the ORDER; this is the cleanest, least-shaped evidence of syntax and it is robust.

Spatial layout uses the space (with a caveat)

Freezing the head collapses the tape organism from 0.83 to 0.41 (= the one-cell ceiling), n=6 -- it really uses the geometry. CAVEAT: this rests on a DESIGNED per-position reward; under pure world-consequence fitness it was unreachable. Report as reward-dependent.

Figure 2. Ablations that held up, and the (refined) delay-invariance



4. What was repaired (artifacts, not lies, but wrong as stated)

'Sequential vastly beats the bag' -- a TRAINING ARTIFACT

Claim: a sequential channel (0.24) beat an order-free bag (0.14), proving the world REQUIRES order.

Hole: both were trained only 600 generations, and the bag converges slower. Its analytic ceiling (convey action + entity-set, then GUESS which entity is the agent = $\times 0.5$) is ~ 0.4 - 0.5 , nowhere near 0.14.

```
bag full-event @ 600 gens : 0.166      (undertrained -- the original number)
bag full-event @ 1500 gens : 0.252      (its real plateau, = action+entities, guess roles)
FAIR (both @1500): sequential full 0.369+/-0.035 vs bag 0.247+/-0.028
                    sequential swap 0.345+/-0.035 vs bag 0.252+/-0.038
```

Repaired verdict: the order advantage is REAL and statistically separated (~ 3 sigma) but MODEST (~ 0.12), not the dramatic gap first reported. The bag does about as well as a role-guesser should. The strong, clean evidence for order remains the SCRAMBLE test, not the bag comparison.

'External memory never forgets' -- REFINED

2-seed sweep gave a noisy 0.48-0.76 'no decay' story. 5-seed sweep (Figure 2, bottom) shows the true shape: a single drop ($L=2$ 0.85 $\rightarrow$ $L=4$ 0.63) then a DEAD-FLAT plateau to $L=24$ (0.61-0.65). The plateau over 20 extra delay steps with zero decay genuinely is the external-storage signature -- a decaying internal memory would keep falling. So the claim is upheld for $L \geq 4$, but at ~ 0.6 (not 0.75) and with an honest initial-difficulty drop.

5. Failures, in painful detail

Content-addressing -- a genuine wall (find by WHAT, not where)

A world that DEMANDS associative retrieval was built: key->value pairs in random arrival order, so position cannot be the address. Three escalating attempts, all failed:

```
linear policy      : ~0.44  -- degenerate fixed guess; a linear map over separate one-hots
                        cannot even compute 'move to the cell indexed by the key'
+ nonlinear hidden : ~0.31  -- now expressible, still converges to a fixed output per
                        query, ignoring storage entirely
+ large value space : ~0.10  -- reaches 'store ONE item, sometimes retrieve it', never
                        two items keyed by content. Ablation: no use of the head.
```

Honest: associative / content-addressed memory is beyond this gradient-free substrate. The capability hierarchy is real -- external storage (reached) < spatial layout (reached, small N) < content-addressing (NOT reached). And the third attempt ('large V to open a gradient') was itself a reward-shaping move, now disavowed. We did NOT keep adding architecture, because 'fix it with a bigger brain' is the gradient-thinking trap.

The foraging 'relocating' control was CONFOUNDED

Intended as 'memory useless -> marks should not help'. But relocating food reappears fresh at new cells, so it is MORE abundant (5.33) than persistent food (4.22). The control changed abundance, not memory's usefulness -- invalid. The valid evidence is the same-organism marks ablation (4.01 vs 1.93) within the persistent world.

Compositionality (D) is near-tautological

Zero-shot held-out generalisation (0.88-1.00) appears ONLY with the anchor, which grounds each word INDEPENDENTLY. Independent per-word grounding IS compositional structure, so recombining into novel phrases is largely guaranteed by construction -- not evidence that compositionality EVOLVED. Honest claim: grounding ENABLES compositional reuse.

Binding (G) leaned on a switched reward

The binding result only appeared after I changed the reward from per-component to per-entity -- i.e. tuned the reward until the result showed. Downgraded to suggestive / reward-dependent, not established. Absolute accuracy was ~0.03 regardless.

6. Capability hierarchy by evidence quality, and the refined thesis

Graded by how clean the evidence is (not how exciting the claim)

external memory (marks/world)	STRONG	natural fitness; ablation 0.67/0.16 & food 4.0/1.9
sex stabilises language	STRONG	n=8; 0.79 vs 0.36; variance mechanism matches
word order is used	MODERATE	scramble 0.22->0.055 (n=8); bag gap modest (fair)
spatial layout	MODERATE	freeze ablation clean BUT designed per-pos reward
constituent binding	WEAK	low absolute; reward was switched-to (shaped)
content-addressing	WALL	fails three ways; not reached

The refined thesis

1. Capabilities emerge from worlds whose survival PHYSICS make them the cheapest way to live. External memory is the clean demonstration: it pays in food, ablation-proven, with zero reward shaping.
2. The reachability 'lever' is NOT a free gradient. Emergence requires that partial progress carries partial survival value -- which real worlds provide (partial communication partially helps; more food is more life) but which I must not COUNTERFEIT with a designed slope. Where I counterfeited it (binding, the proximity probe, large-V), the results are weak or void.
3. There is a real capability ceiling for this gradient-free substrate, and it sits at ASSOCIATIVE (content-addressed) memory. External and spatial memory are reachable; key->value association is not.

What honestly remains to do

- Re-derive binding and spatial layout from a WORLD-CONSEQUENCE fitness (e.g. embodied), and see if they survive without the designed grading.
- A genuinely reachable world for associative memory (embodied, where returning to a remembered place pays in food) -- WITHOUT a proximity slope. Hard, open.
- Confidence intervals on everything still at $n \leq 3$ (the binding/content numbers).

Bottom line

The spine of the arc survives: worlds that demand a capability can evolve it, gradient-free, and external memory shows it cleanly in survival currency. But roughly a third of the headline drama was either a training artifact (the bag) or rested on a reward I shaped (binding, spatial). Stated honestly, the result is narrower and more trustworthy: build the world's physics, never the reward's slope.